Fuse: Performance Evaluation Against NumPy and PyTorch

Platform: Apple M-series · macOS · Single-threaded CPU · Compiled: LLVM OrcJIT, -O3, auto-vectorization · Python 3.13 · April 2026

ABSTRACT

Fuse is an ahead-of-time ML compiler that lowers computation graphs to LLVM IR, applies operator fusion, and executes via OrcJIT. This report benchmarks Fuse against NumPy (1.26) and PyTorch (2.x) on three workloads representative of small-batch CPU inference: a 4-layer MLP, element-wise chains of varying length, and peak intermediate memory during a 5-op transcendental chain. On Apple Silicon, transcendental operations (`sigmoid` , `tanh` , `exp` , `log`) are dispatched through Apple's vForce library (`vvtanhf` , `vvexpf`) via the `VecLibSplitPass` . Fuse is 1.93× faster than NumPy and 2.09× faster than PyTorch eager on MLP inference, and requires 3× less peak intermediate memory than NumPy and 2× less than PyTorch on a 5-op chain.

§1 MLP Inference Latency

Configuration: 4 fully-connected layers, 256 hidden units, ReLU activation, batch size 1. Latency is mean over 10,000 forward passes (µs). Lower is better.

System	Latency (µs)	Ratio vs Fuse	Primary bottleneck
Fuse	**5.2**	—	BLAS GEMM + fused JIT epilogue; zero runtime allocation
NumPy	10.1	1.93×	3 separate kernel passes; 3 heap allocations per inference
PyTorch eager	10.9	2.09×	Per-op Python dispatch (~2 µs each); unfused bias and activation
torch.compile	17.2	3.29×	Compilation overhead dominates at batch=1 with only 4 small layers

The MLP is a `MatMul → BiasAdd → ReLU` sequence repeated four times. Fuse fuses BiasAdd and ReLU into the matmul epilogue via the `MatmulEpiloguePass` , invoking BLAS for the GEMM core and a single JIT-compiled loop for the trailing activations. NumPy and PyTorch each dispatch these as three independent operations, incurring

separate kernel launches and, for NumPy, separate heap allocations. At batch size 1, torch.compile's fixed dispatch cost exceeds any fusion benefit it provides.

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§2 Element-wise Chain — N = 1,000,000

Chain: `add → relu → mul → sigmoid → tanh` over 1M float32 elements. Latency is mean over 1,000 runs (ms). Lower is better.

System	Latency (ms)	Ratio vs Fuse	Mechanism
Fuse	**1.39**	—	Fused LLVM JIT (add/relu/mul) + Apple vForce (sigmoid/tanh)
PyTorch eager	1.52	1.09×	5 separate ATen kernels; `exp` via Intel MKL (ARM path slower)
NumPy	2.69	1.94×	5 unfused loops; no vecLib dispatch; 5 intermediate allocations

The non-transcendental prefix (`add` , `relu` , `mul`) is handled by the fused LLVM JIT kernel in a single pass. The `VecLibSplitPass` extracts `sigmoid` and `tanh` before fusion and routes them to Apple's vForce library (`vDSP_vneg` + `vvexpf` + `vDSP_vsadd` + `vDSP_svdiv` for sigmoid; `vvtanhf` for tanh). These are hand-tuned ARM NEON routines that operate on entire arrays rather than element-by-element through a scalar libm loop.

> *On Linux, `VecLibSplitPass` is a no-op. Transcendental chains fall back to LLVM's generic `exp()` intrinsic and may not beat PyTorch's MKL path on x86.*

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§3 Fusion Benefit vs Chain Length

Raw latency (ms) and Fuse speedup ratios vs NumPy and PyTorch as chain length increases, measured at two array sizes. The two regimes behave differently: at N = 10K, data fits in L1/L2 cache and per-op dispatch overhead dominates; at N = 5M, each array is 20 MB and memory bandwidth dominates.

N = 10,000 — cache-resident, dispatch-dominated

Chain length	NumPy (µs)	PyTorch (µs)	Fuse (µs)	vs NumPy	vs PyTorch
1 op	3	2	4	0.73×	0.60×
2 ops	7	5	4	1.75×	1.22×
3 ops	9	7	4	2.12×	1.64×
5 ops	48	42	15	3.12×	2.75×
8 ops	131	189	27	4.95×	**7.13×**

N = 5,000,000 — bandwidth-dominated

Chain length	NumPy (ms)	PyTorch (ms)	Fuse (ms)	vs NumPy	vs PyTorch
1 op	1.150	0.639	1.771	0.65×	0.36×
2 ops	2.472	1.129	1.755	1.41×	0.64×
3 ops	3.152	1.780	1.770	1.78×	1.01×
5 ops	14.858	8.362	8.326	1.78×	1.00×
8 ops	25.417	11.974	13.778	1.84×	0.87×

The two regimes tell different stories. At N = 10K, PyTorch's per-op dispatch overhead accumulates faster than NumPy's as chains grow — at 8 ops, PyTorch (189 µs) is actually *slower* than NumPy (131 µs), while Fuse (27 µs) is 7.13× faster than PyTorch. Fusion eliminates both the kernel-launch overhead and the intermediate buffer passes in a single fused loop.

At N = 5M, the picture changes. PyTorch consistently beats NumPy because its ATen kernels use highly optimized SIMD (MKL / Accelerate) for each operation individually, and at bandwidth-saturated sizes that per-op SIMD quality matters more than the number of passes. Fuse ties PyTorch at 3–5 ops and falls slightly behind at 8 ops, where PyTorch's per-op Accelerate routines outperform Fuse's fused loop for expensive transcendentals (`exp` , `sqrt`) at very large N. Fuse still outperforms NumPy by 1.8× at all chain lengths ≥ 3.

> *Fuse's fusion advantage is largest in the cache-resident regime (small-to-medium N) where dispatch overhead dominates. At very large N, where bandwidth is the ceiling, PyTorch's per-op vectorized math libraries are competitive. The crossover vs PyTorch occurs around N = 500K for chains without transcendentals and N = 1M–5M for chains that include them.*

§4 Peak Intermediate Memory

Peak intermediate memory allocated during execution of the 5-op chain (`add → relu → mul → sigmoid → tanh`) over N = 1,000,000 float32 elements. Inputs are pre-allocated and excluded; only allocations caused by chain

execution are counted.

NumPy	12.0 MB
PyTorch	8.0 MB
Fuse	4.0 MB

System	Peak memory	Ratio vs Fuse	Mechanism
NumPy	12.0 MB	3.0×	One `malloc` per operation; up to 3 buffers live simultaneously
PyTorch	8.0 MB	2.0×	Caching allocator reuses freed blocks, but old and new tensors coexist during each assignment
Fuse	**4.0 MB**	**—**	2 buffer slots pre-allocated at compile time; ping-ponged via `BufferReusePass`

Each element in the chain is a 4 MB array (1M × float32). NumPy allocates a new output array per operation; with CPython reference counting, up to three arrays can be simultaneously live (input, current output, next-op input overlap). PyTorch's `c10` caching allocator avoids repeated `malloc` calls but cannot eliminate the transient double-buffer window: when executing `x = op(x)`, the old and new tensors overlap until the assignment completes, yielding a peak of 2 × 4 MB = 8 MB. Fuse's `BufferReusePass` performs liveness analysis at compile time and assigns only two logical buffer slots, ping-ponging between them throughout the chain — 4 MB regardless of chain length.

> *Measurement methodology: NumPy and Fuse are measured via `tracemalloc` (CPython allocator hooks). PyTorch's `c10::Allocator` bypasses the Python allocator entirely, so it is measured by tracking `tensor.nbytes` at each step of the chain and recording the maximum simultaneously-live byte count.*

§5 Summary

The following table aggregates results across all three workloads for the primary comparison target: small-batch CPU inference on Apple Silicon.

Metric	Fuse	NumPy	PyTorch eager	torch.compile
MLP latency (µs)	**5.2**	10.1	10.9	17.2
Element-wise chain (ms)	**1.39**	2.69	1.52	—
Peak memory (MB)	**4.0**	12.0	8.0	8.0
Heap allocs per inference	**0**	3–5	3–5	3–5

Metric	Fuse	NumPy	PyTorch eager	torch.compile
Supports GPU	No	No	Yes	Yes
Supports arbitrary ops	No (element-wise + matmul)	Yes	Yes	Yes

Fuse's advantage is specific and bounded. It wins on any workload expressible as element-wise chains and matmul with activations — the dominant patterns in feed-forward inference at small batch sizes. It does not address large-batch GEMM (where BLAS already saturates throughput), convolutions, attention, or any operation outside its supported op set. On Linux or x86, the vForce advantage for transcendentals is absent and the margin over PyTorch narrows.

The performance derives from three compounding mechanisms: (1) operator fusion eliminates intermediate buffer passes over the data, amortizing the memory bandwidth cost across the full chain; (2) the `BufferReusePass` reduces peak allocation to a fixed 2-slot pool regardless of chain length; and (3) on Apple Silicon, the `VecLibSplitPass` routes transcendental functions through vForce, replacing LLVM's scalar libm loop with ARM-native SIMD implementations.